DEADLINE -27/1/2026 at 5:30pm.
LINK: https://forms.gle/ty96M2djFCnmcQD48

WEIGHTAGE - 5 MARKS,

- ONE GENERATIVE AI ANSWER CAN COST -5 STRAIGHT MARKS.
- PALAGIARISM IS NOT ALLOWED(NEITHER THROUGH FRIENDS NOR THROUGH WEBSITE). IF FOUND THEN STUDENT WHO SHOWN AND STUDENTS WHO HAVE SEEN WILL GET ZERO WITHOUT FURTHER ARGUMENTS.
- ALL THE CODES SHOULD BE UNIQUE

The Below Question has to be answered in a paper.

Assignment

Q: Read at most 10 names of students and store them into an array of String nameOfStudents[10]. Sort the names into the lexicographic order. Display the sorted list of names.
Q: Define a class Complex to represent an object for a complex number like Z = X + i.Y with the following methods:

```
Complex add(Complex z1, Complex z2) //To add two complex numbers
Complex sub(Complex z1, Complex z2) //To subtract two complex numbers
Complex mul(Complex z1, Complex z2) // To multiply two complex numbers
float magnitude(Complex z)            // To find the modulus
Complex conjugate(Complex z)       // To find the complex conjugate
```

Write the main class and instantiate the objects of the above mentioned classes.
Q: Add the necessary methods in the class PointCreate3 (Practice 3.8) to calculate the area and perimeter of a rectangle given the two corner coordinates.
Q: Read at most 10 names of students and store them into an array of String nameOfStudents[10]. Sort the names into the lexicographic order. Display the sorted list of names.
Q: Define a class Employee with usual member for an employee like empCode(String), empName(String), dateOfBirth(Date), dateOfJoin(Date), designationCode(int), salary(float).Create a list to store data about 10 employees using Vector. Manipulate the list using the methods in class Vector.
Q: Define an one dimensional array "vivid" of type float. Read the values from the keyboard to load the array. Calcualte and then print the average of all the values in "vivid"..
Q: Define two 2D arrays of integers, namely A[3]4] and B[4][2]. Store the values into them. Store the result of matrix multiplication into an another 2D array, say C.
Q: Write a program to store a lists of name in a List. Reverse the order of the names in the list using Stack. You should use class ArrayList to store the names (String) and class Stack for reversing..
Q: Create a class ArrSort which contains one double array to store the data and one integer variable to store the number of elements.
Q: Write a program to read the content of a file and count the number of words in the file.
Q: Write a program to copy contents from one file to another file.
Q: Write a program to merge two file contents into another file.

Q: Define two different classes namely, Student and Employee. These classes are derived from a base class Person. Define other two classes Staff and Faculty. Staff and Faculty classes are derived from Employee class. The Person class has name and age data and display method to display the name and age of a person. The Student class has data like rollNo and branch and display method to display name, age, rollNo and branch of the student. Stuff has ecNo and doj(date of joining) data and display method to display name, age, ecNo, doj of the stuff. Faculty has designation data (Assistant Professor, Associate Professor and Professor) and display method to display the name, age, ecNo, doj and designation of the Faculty. Staff has designation data (Technical and Clerical) and display method to display the name, age, ecNo, doj and designation of the Staff. Each class have their own constructor to initialize the value of each data field. Finally create MainDemoClass and create an object of each class. Print the values of all objects in the MainDemoClass.

```
Practice 3.8
/* Automatic initialization - concept of constructor */
        class Point ()  {
        int x, y;

        Point ( int x, int y ) {
        this.x = x ;
        this.y = y;
        }

        printPoint() {
        System.out.println("X = "+ this.x + " Y= " + this.y);
        }
        }

        class PointCreate2 {
        public static void main ( String args [  ] ) {
        Point p = new Point (10, 20 );
        p.printPoint();
        }
        }
```

FOR THE BELOW QUESTION CODE NEED TO BE SUBMITTED:

CREATE A JAVA PACKAGE mathstud THAT CONTAIN JAVA CODE FOR THE BELOW(Do not use an inbuilt library):

- Find LCM of 2 numbers
- Find GCD or HCF of 2 numbers
- Display All Prime Numbers from 1 to N
- Check Leap Year
- Check Armstrong Number between Two Integers
- Check whether the input number is a Neon Number
- Check whether input character is vowel or consonant
- Find Factorial of a number
- Find Even Sum of Fibonacci Series Till number N

CREATE A JAVA PACKAGE FOR SORTING THAT CONTAINS THE FOLLOWING SORT (NO need of using inbuilt library)

- RADIX SORT (only through array/matrix)
- BUCKET SORT (only through array/matrix)
- HEAP SORT
- BUBBLE SORT
- SELECTION SORT
- INSERTION SORT
- CREATION OF STACK and QUEUES WITH ALL OPERATIONS.